

Utkarsh Singh

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ABOUT

Computer Science graduate with a strong foundation in software development, problem-solving, and modern web technologies. Experienced in designing and developing scalable applications, RESTful APIs, and database-driven solutions. Quick learner with strong analytical skills, a collaborative mindset, and a passion for building efficient and reliable software systems.

EDUCATION

• Babu Banarasi Das University	Lucknow, India
• <i>Bachelor of Technology in Computer Science and Engineering; CGPA: 7.27</i>	2021 – 2025
• Sant Atulanand Convent School	Varanasi, India
• <i>Senior Secondary Education; Percentage: 74.2%</i>	2021
• Udai Pratap Public School	Varanasi, India
• <i>Secondary Education; Percentage: 62.83%</i>	2019

SKILLS SUMMARY

- **Programming Languages:** Python, JavaScript, HTML, CSS
- **Frontend Development:** React.js, Tailwind CSS, Responsive Web Design
- **Backend Development:** FastAPI, Flask, RESTful APIs, API Integration
- **Databases:** MySQL
- **AI and Machine Learning:** OpenCV, LangChain, ImageHash
- **Libraries and Tools:** Pandas, FAISS, NumPy, Hugging Face Embeddings, Git, Google Gemini API

EXPERIENCE

• Four Corners Technologies Pvt. Ltd	Bengaluru, India
• <i>Tech Intern</i>	April 2025 – October 2025
◦ Backend Development: Developed Python-based solutions for data processing and backend tasks.	
◦ Workflow Automation: Automated data workflows using Python scripts to improve operational efficiency.	
◦ Data Validation: Performed data validation and implemented robust error-handling mechanisms to ensure accuracy and reliability of processed data.	
◦ Database Handling: Wrote optimized SQL queries and managed structured data to support backend operations.	
◦ Performance Optimization: Improved script performance by refactoring code and reducing execution time through efficient logic design.	
◦ Collaboration: Collaborated with cross-functional teams to deliver scalable and efficient technical solutions.	

PROJECTS

- **Dataset Cleaner AI** 
 - **AI-Powered Dataset Cleaning:** Developed an intelligent dataset cleaning platform to automatically identify and remove duplicate and blurry images from large-scale image datasets.
 - **Computer Vision Implementation:** Utilized OpenCV and perceptual image hashing techniques to perform image quality assessment and duplicate image detection.
 - **Full-Stack Development:** Built a responsive user interface using React.js and integrated it with a FastAPI backend for image upload, processing, and visualization.
 - **Workflow Automation:** Streamlined dataset preprocessing workflows, reducing manual effort and improving data quality for machine learning applications.
 - **Tech Stack:** React.js, FastAPI, Python, OpenCV, ImageHash, Tailwind CSS.
- **Retrieval-Augmented Generation (RAG) System** 
 - **RAG Pipeline Development:** Built a Retrieval-Augmented Generation system capable of answering user queries using custom PDF, TXT, and Markdown documents.
 - **Semantic Search Engine:** Implemented document chunking, embedding generation, and FAISS vector storage for efficient semantic retrieval of relevant information.
 - **LLM Integration:** Integrated Google Gemini with LangChain RetrievalQA to generate context-aware responses grounded in retrieved content.
 - **Response Optimization:** Enhanced answer relevance and reduced hallucinations through retrieval-based grounding and context injection techniques.
 - **Tech Stack:** Python, LangChain, Google Gemini API, FAISS, Hugging Face Embeddings.

CERTIFICATIONS

- : Python Programming – Shri Balaji Infotech
- : MySQL – HackerRank
- : HTML – Great Learning
- : Python – Great Learning